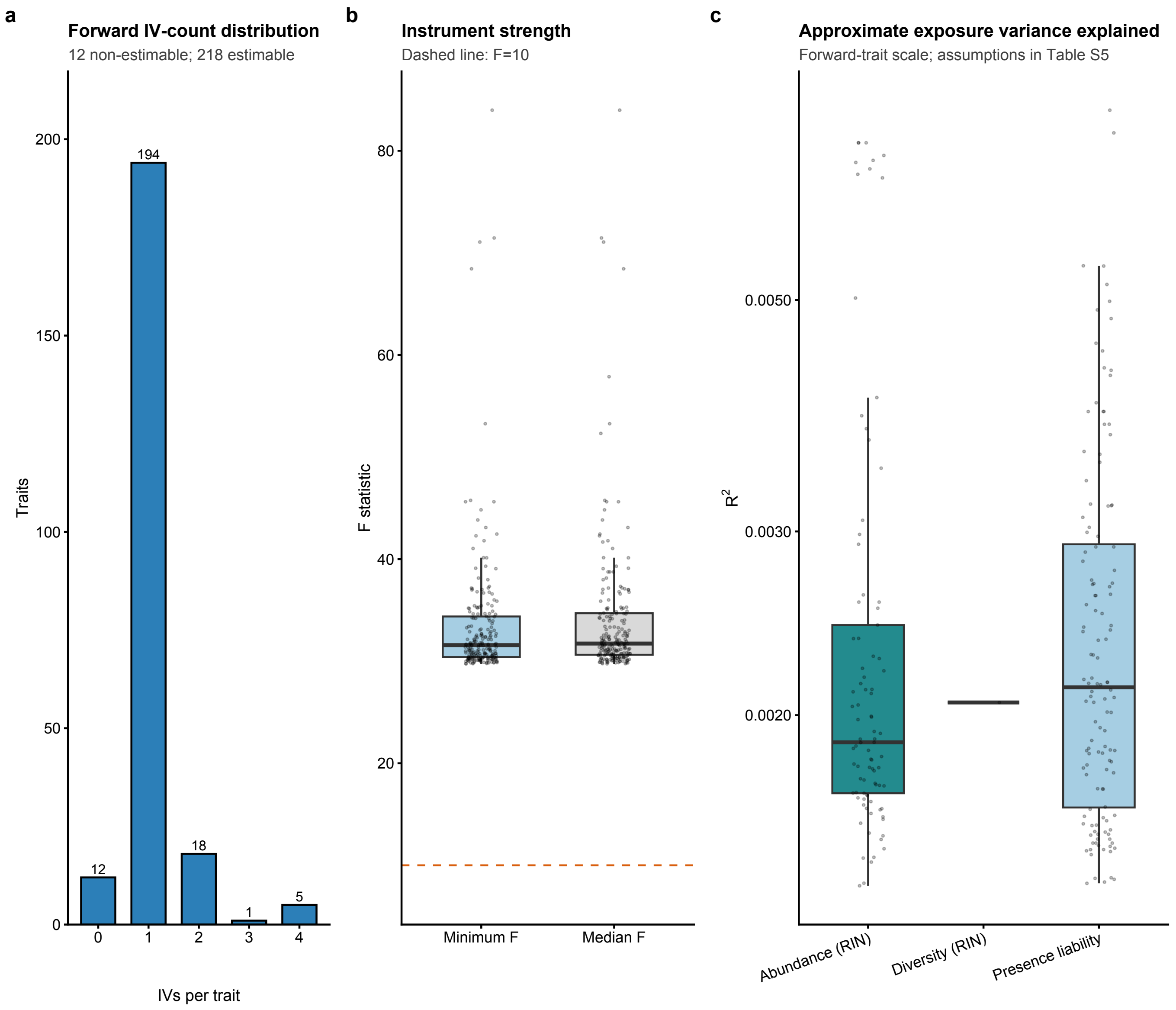
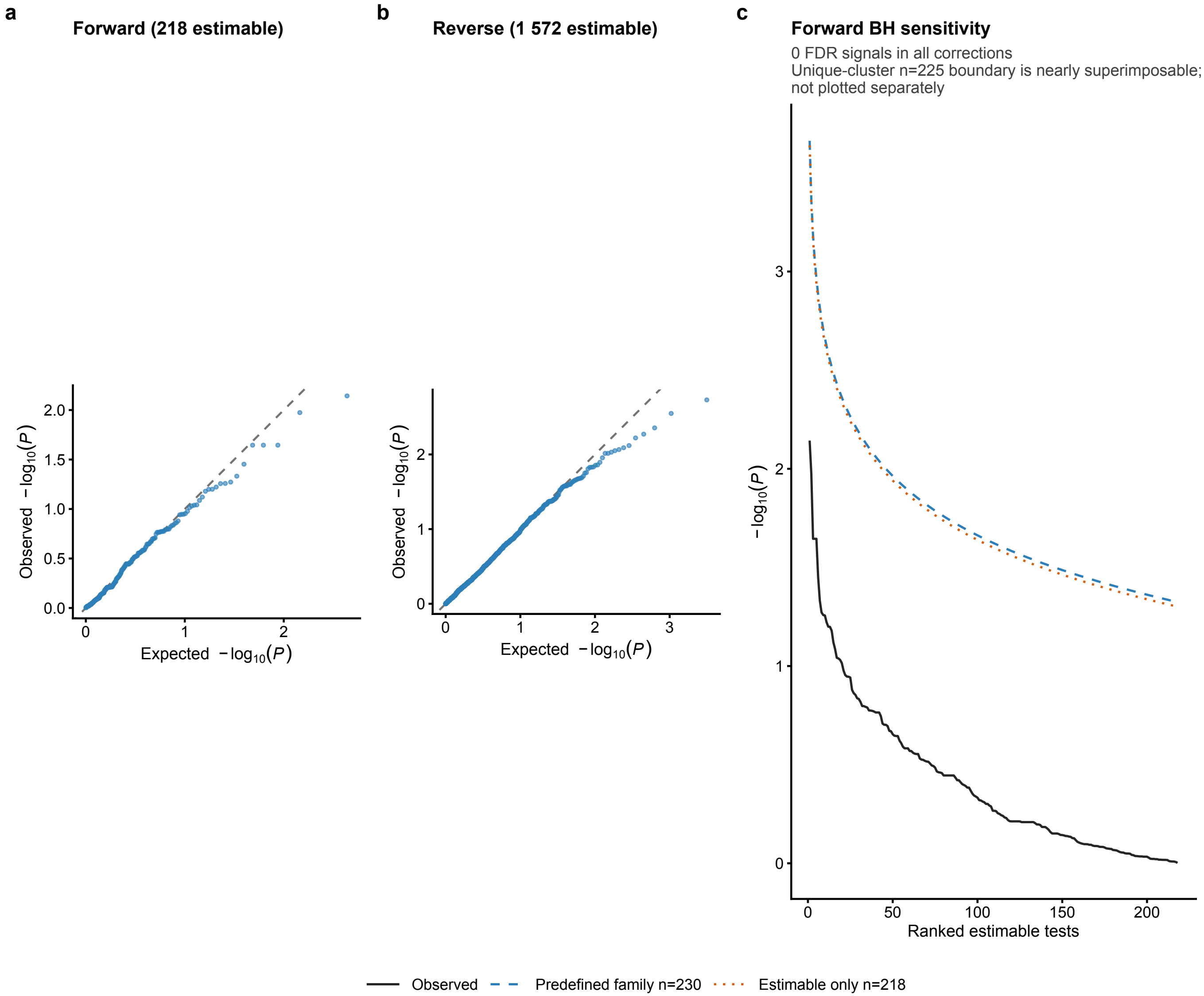
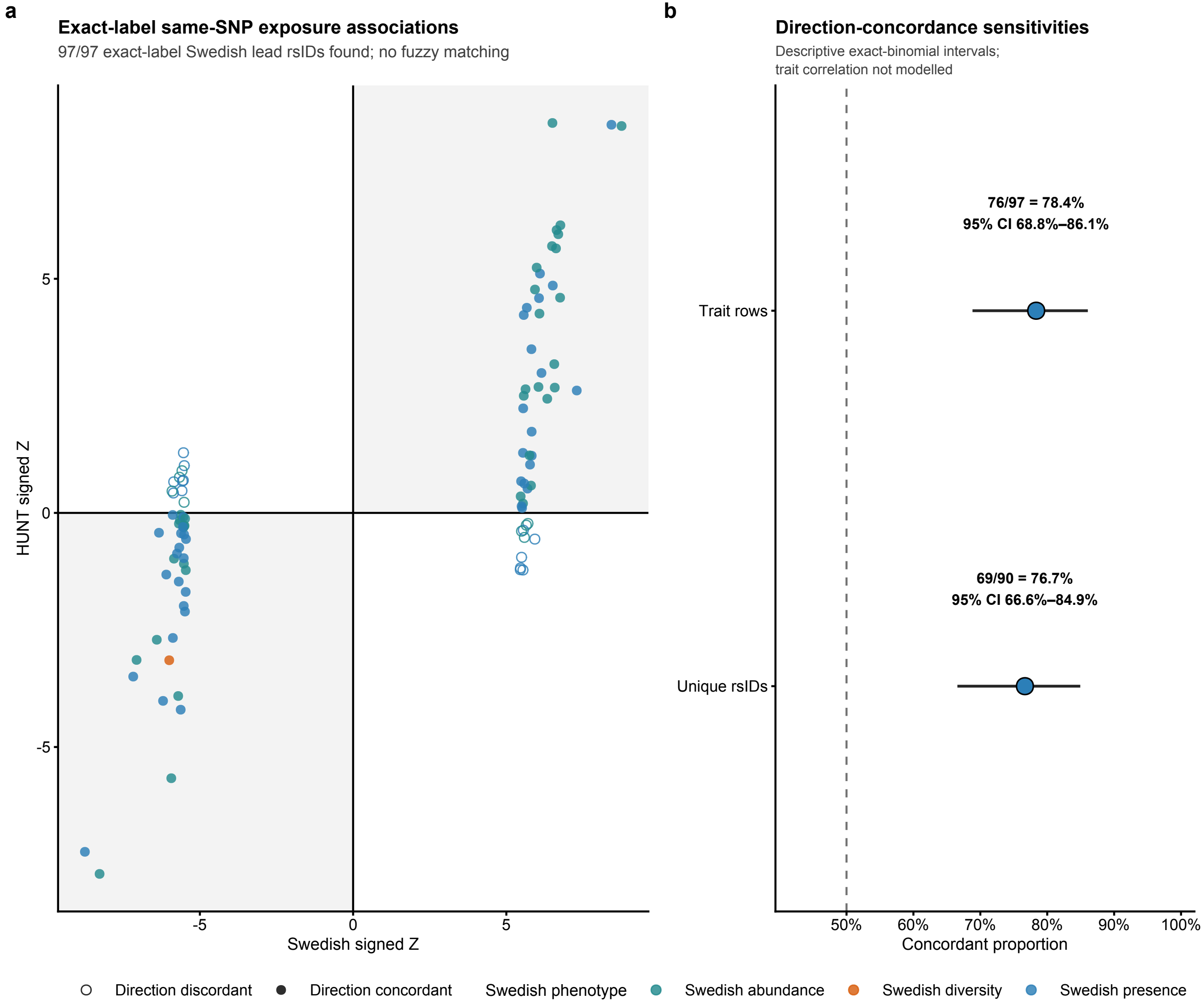
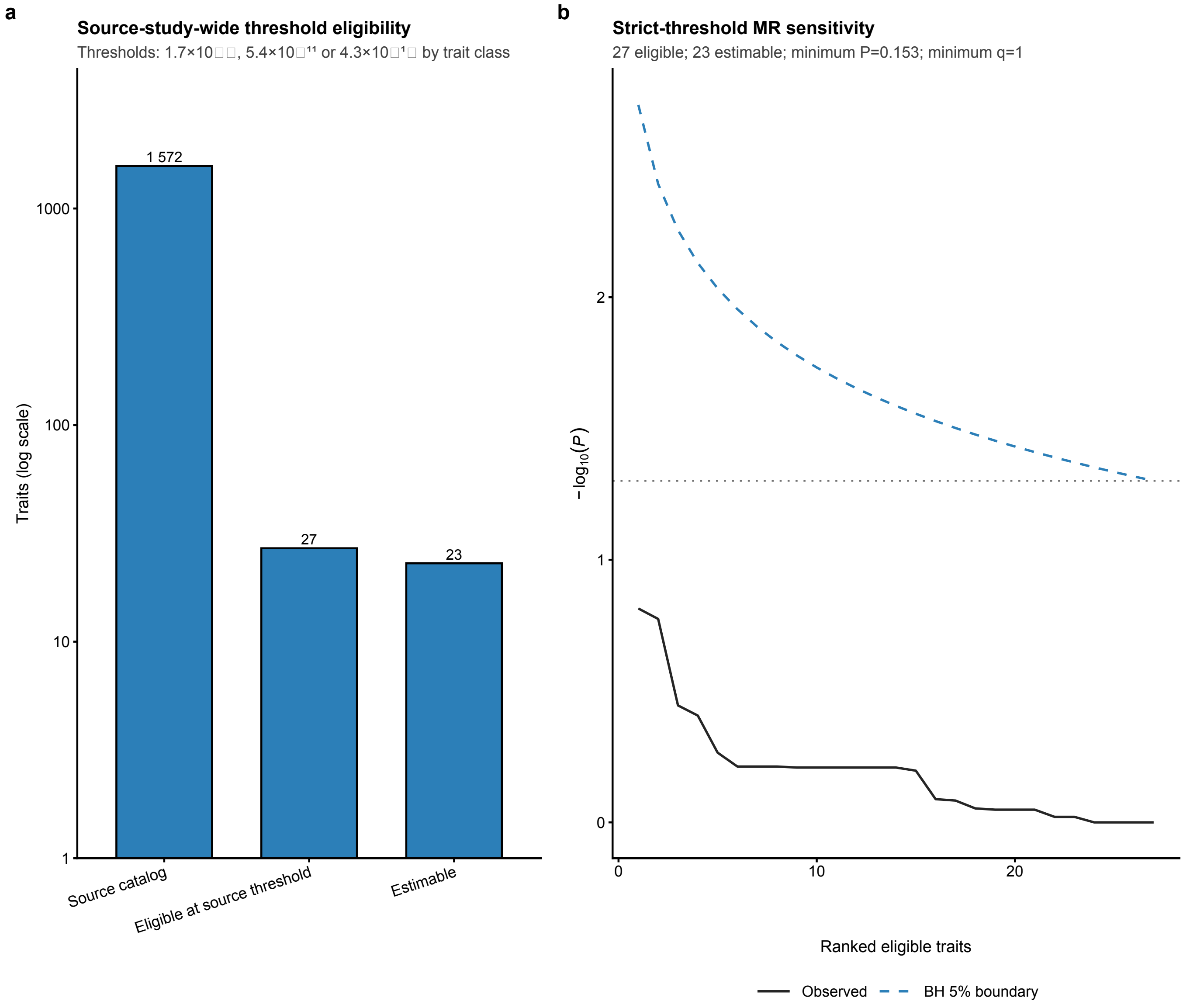

a
Forward IV-count distribution
12 non-estimable; 218 estimable
Traits
IVs per trait
194
18
12
1
5
0
50
100
150
200
0
1
2
3
4
b
Instrument strength
Dashed line: F=10
F statistic
20
40
60
80
Minimum F
Median F
c
Approximate exposure variance explained
Forward-trait scale; assumptions in Table S5
R^2
0.0020
0.0030
0.0050
Abundance (RIN)
Diversity (RIN)
Presence liability

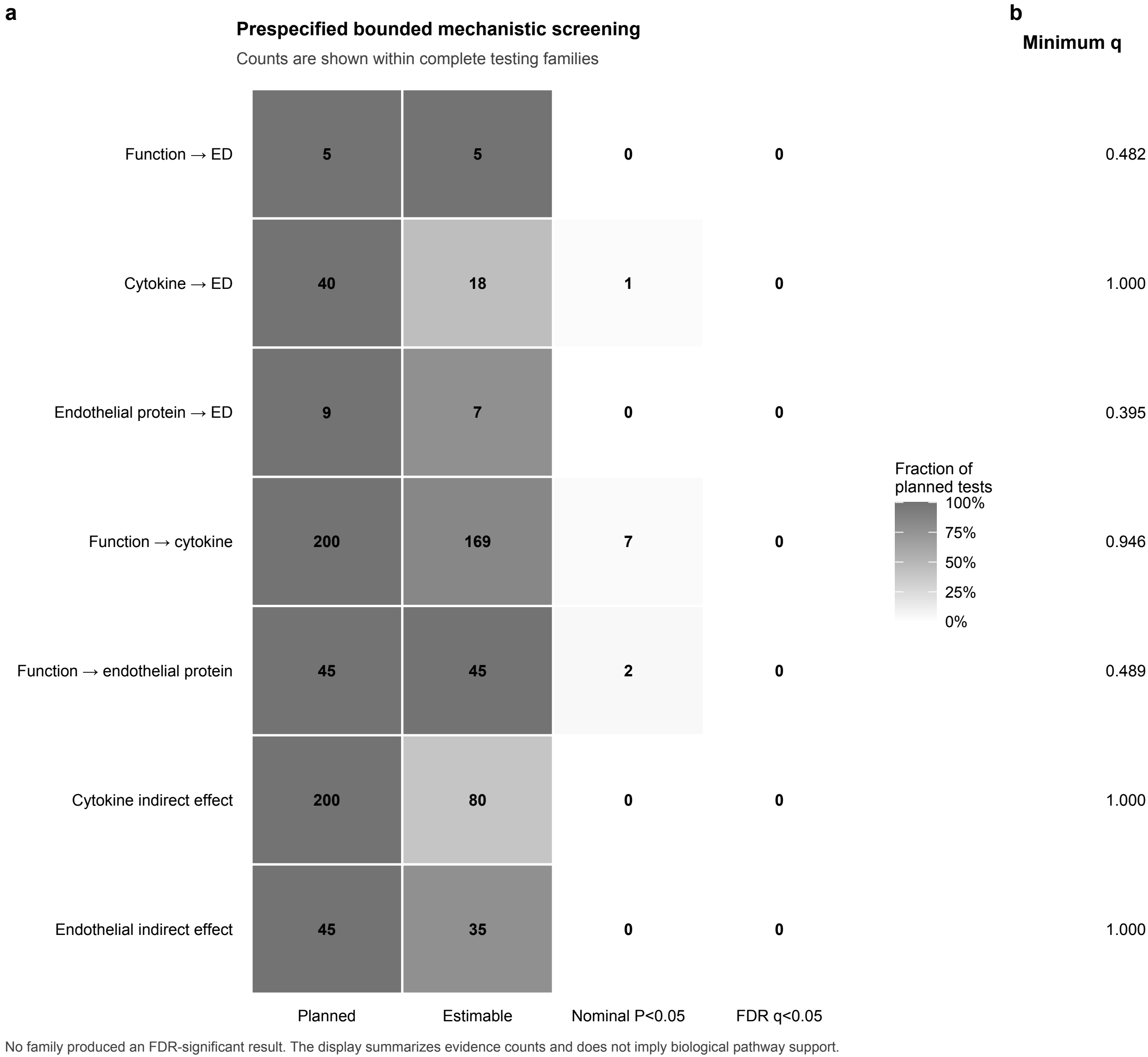


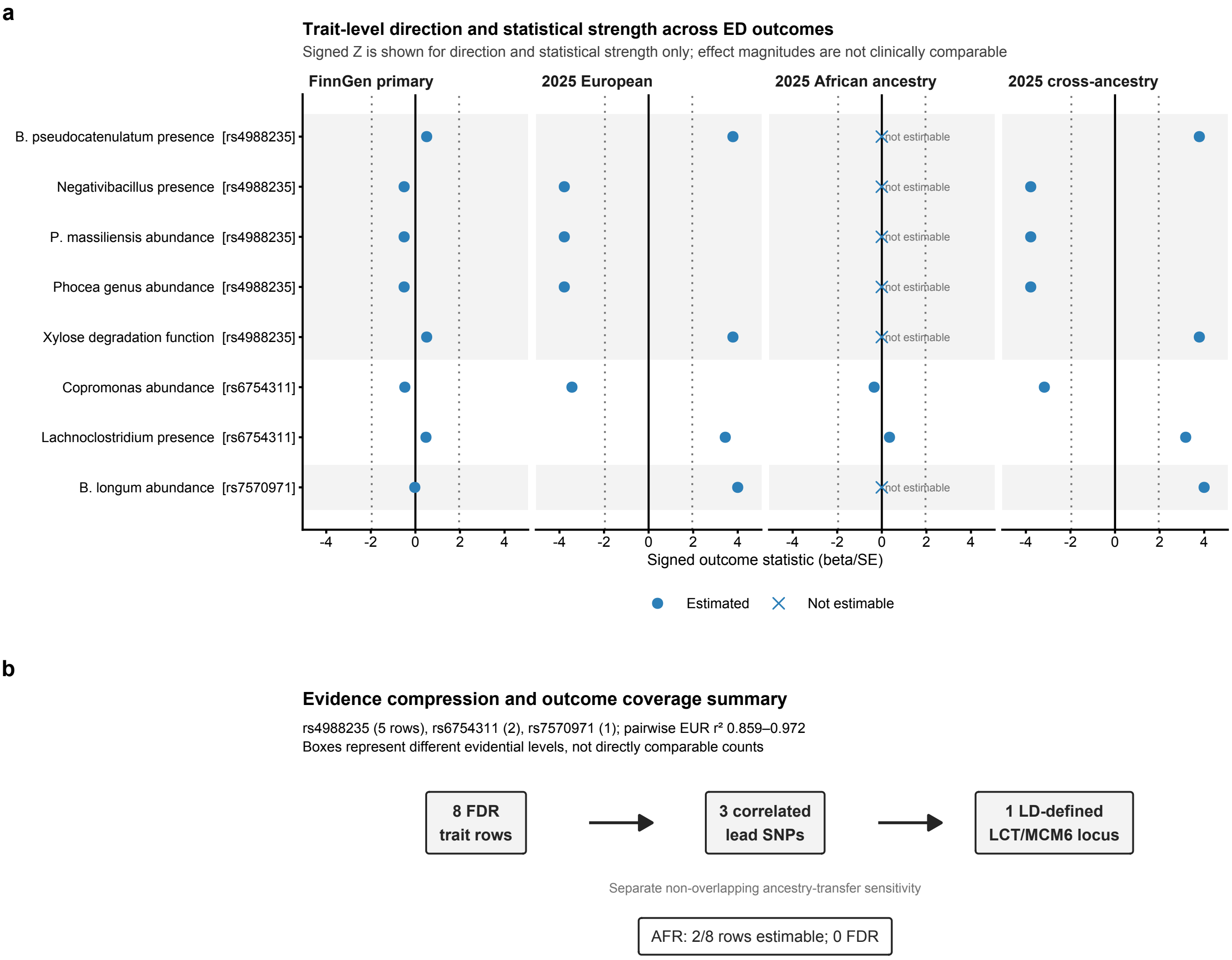
QQ plots are descriptive because microbial traits are correlated; no genomic-inflation estimate is interpreted as if tests were independent.



Signed Z supports a direction-only comparison. Swedish presence and HUNT normalized abundance scales preclude direct comparison of raw beta or MR OR magnitude.


a
Source-study-wide threshold eligibility
Thresholds: 1.7×10□□, 5.4×10□¹¹ or 4.3×10□¹□ by trait class
Traits (log scale)
1
10
100
1000
1 572
27
23
Source catalog
Eligible at source threshold
Estimable
b
Strict-threshold MR sensitivity
27 eligible; 23 estimable; minimum P=0.153; minimum q=1
$-\log_{10}(P)$
0
1
2
0
10
20
Ranked eligible traits
Observed
BH 5% boundary


a
Prespecified bounded mechanistic screening
Counts are shown within complete testing families
Function → ED
Cytokine → ED
Endothelial protein → ED
Function → cytokine
Function → endothelial protein
Cytokine indirect effect
Endothelial indirect effect
5 5 0 0
40 18 1 0
9 7 0 0
200 169 7 0
45 45 2 0
200 80 0 0
45 35 0 0
Planned
Estimable
Nominal P<0.05
FDR q<0.05
Fraction of planned tests
100%
75%
50%
25%
0%
b
Minimum q
0.482
1.000
0.395
0.946
0.489
1.000
1.000
No family produced an FDR-significant result. The display summarizes evidence counts and does not imply biological pathway support.



The eight European and cross-ancestry trait-level associations surviving FDR correction are not eight independent biological findings.
The three lead SNPs form one high-LD locus; African non-estimability reflects SNP coverage rather than evidence of absence.